Queen Mary University
EECS — Music Perception and Cognition

Brief Auditory Storage

Take the “Darwin1972” and “Treisman1972” as examples

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[DTC72]: An Auditory Analogue of the Sperling Partial Report Procedure: Evidence for Brief Auditory Storage

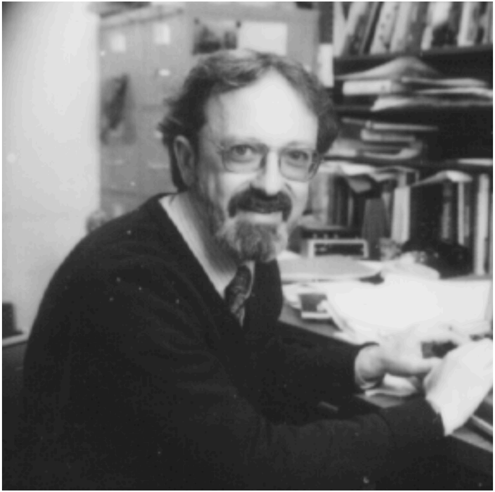


Figure: George Sperling (1934-)

Purpose of 3 Experiments

1. Explore the effect of time delay on PR and of requiring report by spatial location and/or by letter/digit category.
2. Participants: 12 undergraduates

Design of 3 Experiments

Index	Type	Location	Category	Method
Exp1	WR	✓		
	PR	✓		with visual indicator
Exp2	WR	✓	✓	with visual indicator & order requied
	PR	✓		order not requied
Exp3	WR		✓	
	PR		✓	with visual indicator

Stimuli

```
# 2 categories each 9 elements
number_list[9] =  [1, 2, 3, 4, 5, 6, 8, 9, 10]
letter_list[9] = [B, F, J, L, M, Q, R, U, Y]
# 3 spatial locations
spatial[3] = [left_ear, middle_ear, right_ear]
# 4 delays (seconds)
# use different tones to trigger
delay[4] = [0, 1, 2, 4]
```

- This gave a set of 20 main stimuli, each of which had 3 lists of 3 items.
- number of item lists: stimuli[20] $\times$ item[3] $\rightarrow$ 60 item lists
- number of elements: stimuli[20] $\times$ item[3] $\times$ 3 elements $\rightarrow$ 180 elements

Stimuli: (20×9 matrix)

index of stimuli[20]	**item[3]**	**item[3]**	**item[3]**
stimuli[1]	[___,___,___]	[___,___,___]	[___,___,___]
stimuli[2]	[___,___,___]	[___,___,___]	[___,___,___]
stimuli[3]	[___,___,___]	[___,___,___]	[___,___,___]

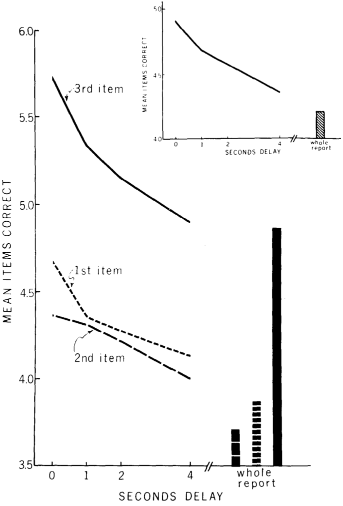
. . . . .

index of stimuli[i]	**item[3]**	**item[3]**	**item[3]**
stimuli[20]	[___,___,___]	[___,___,___]	[___,___,___]

Task

1. *WR*: report all items they can remember in correct locations
   - (no visual indicator)
   - subjects respond on their own time
2. *PR*: (every 0,1,2 or 4 seconds) with *visual indicator*: recall 3 items from one of the locations.
   - (with visual indicator)
   - a slide with blank bar on the left, middle or right

Results



- superiority of WR & PR?
- recency effect?
- a memory store of 2s?

Results

- Recency Effect
  - The 3rd item was recalled better than the other two items
- Memory Store?
  - Functionally define a memory store in which material is held for about 2 seconds.
  - not clear: form of preservation material
  - no direct evidence: the store is specific to auditorily presented material

A Lever!

PR is only superior to WR in the visual case when report is cued along some simple dimension of the stimulus.

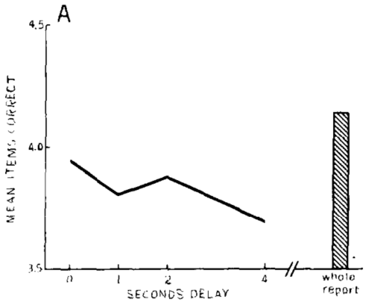
Purpose

1. Does recall by category give an advantage for PR over WR similar to that obtained for recall by spatial location?
2. Participants: 11 undergraduates

Task

1. *WR*: write down in their correct location items of the particular category denoted by the indicator
   - Answer: what were the numbers, and where were they?
   - (required to recall items *in order*)
2. *PR*: to put each item to its correct location
   - (order not required)

Results



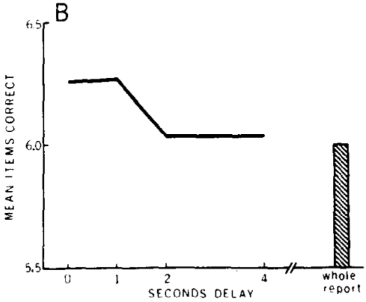
- Report by category is an inefficient mode of recall
- no advantage for PR or effect of delay
- When location was required, PR was lower than WR and showed no significant decay with delay of the partial report indicator

Task

Subjects were not told to remember or report the location of a particular item.

- WR: wrote the numbers on the left and the letters on the right
- PR: wrote the cued category in the left column

Results



- PR becomes more efficient
- approximately same level
- When location was not required, PR was superior to WR for indicator delays of less than 2 seconds
- PR was significantly greater than WR for the 0 and 1-sec delay conditions ($p \leq .01$)

results are campatible with a kind of store which has a useful life of around 2 seconds and from which material may be retrieved more *easily* by spatial location than by semantic category.

[TR72]: Brief auditory storage: A modification of Sperling’s paradigm applied to audition

Figure: Anne Treisman (1935-2018)[1]

[1]Photo by Hao-Hsiang You

Purpose

1. Confirm the Brief Auditory Store (BAS)
2. Determine its decay time
3. Participants: 2 undergraduates

Brief Auditory Store

- transient memory
- for relatively uncategorized material
- must be distinguished from *short-term memory* (central & for categorized material).

Use A Better Procedure

a better procedure here to avoid causing the difficulty for subjects to assiociate stimuli reliably with thier locations:

- substituting *partial recognition* for *partial recall*

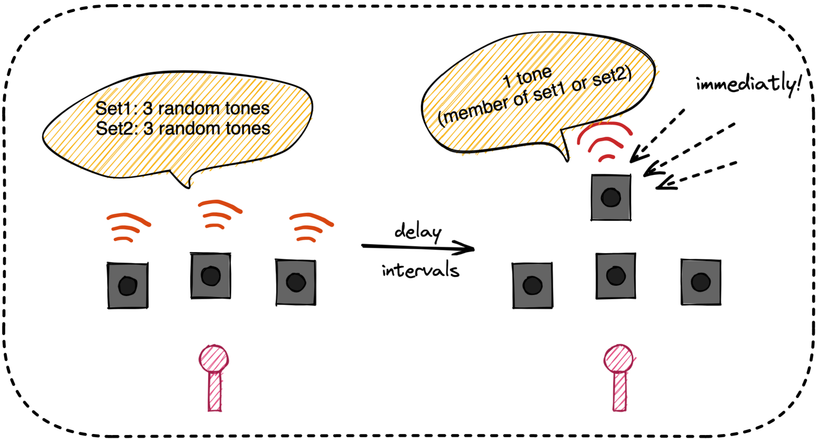
One disadvantage of Darwin1972's experiments!

no attempt to control, as in *Triesman1964*'s experiments[2], the attentional strategy of the subject.

↪ – the General Discussion part of Darwin et al., 1972

[2] Treisman A. Monitoring and storage of irrelevant messages in selective attention[J]. Journal of Verbal Learning and Verbal Behavior, 1964, 3(6): 449-459.

Design of Stimuli



tones[6] = [100, 200, 450, 920, 1800, 6000] (Hz)
delay[4] = [0, 400, 800, 1600] (msec)

Task

1. whether or not the probe had been a member of stimulus set 1
2. whether or not the probe had been a member of stimulus set 2

Results

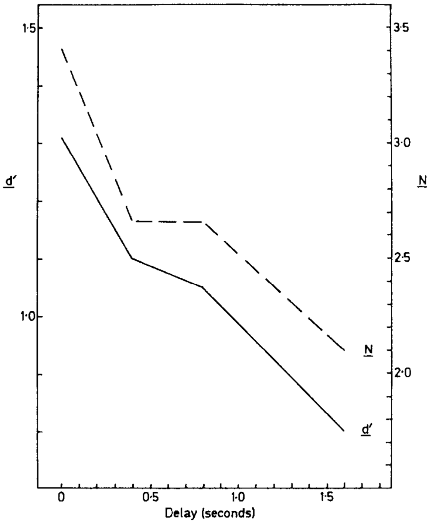
The result can be expressed as a probability form:

- Y: (subjects report) the probe has been previously presented
- N: (subjects report) the probe has not been previously presented
- S: it was included in the given stimulus set on that trail
- $\bar{S}$: it was not included in the given stimulus set on that trail

Example - a conditional probability

$p(Y|\bar{S})$ means the subject report that the probe presented before, *given* it was not included in the stimulus set.

Results



- continuous line: mean value of d' (sensitivity index) for the 2 subjects at the 4 delay intervals
- dashed line: the number of items in store N (calculated from the `loss of items and response guessing` model)

[STB61]: the *signal detectability theory* [3]

- each item was represented by a *decision axis*
- the observation on this axis at time t depending on whether the item was (S) or was not ($\bar{S}$) included in the given stimulus set.
- values of d' and β were calculated
  - d': sensitivity index
    - d-prime: a quantity to define the discriminability of a signal from the background noise[4]
    - when d' is large, the signal will be easier to discriminate from the noise
  - β: Response Bias / Optimum Criterion

[3]Swets J A, Tanner Jr W P, Birdsall T G. Decision processes in perception[J]. Psychological review, 1961, 68(5): 301.

[4]Murphy K P. Machine learning: a probabilistic perspective[M]. MIT press, 2012.

The possible effect of guessing cannot be ignored!

- subjects may have chosen the responses randomly
- (researchers here is rigorous in their control of the uncertainty condition)
- subjects will guess the identity of the probe on the basis that all members of ensemble have been sampled with equal probability, without duplication
- positive response will only result if the item guessed to fill one of the empty locations corresponds to the probe

- at time t, when the probe is presented, there is a probability $p(t)$ that any one item has been retained.

probability of a correct response

$$P(Y \mid S) = p(t) + (1 - p(t))[((1 - p(t))^2/2 + 2p(t)(1 - p(t))(2/5) + p(t)^2/4]$$

- $P(Y|S)$: presented before, given included in set
- $P(Y|\bar{S})$ can also be calculated.
- $p(t)$ can be calculated from $P(Y|S)$
- the number of items N in store: $6p(t)$

implausible...

On these assumptions, $p(t)$ can be calculated from $P(Y|S)$, and the number of items in store at that time can be taken as $6p(t)$. The mean value calculated in this way was 2.84 items at zero delay, falling to 0.63 items at 1.6 sec. But the parameter values obtained from $P(Y|S)$ predict a mean $P(Y|\bar{S})$ of 0.42, whereas the mean experimental value was 0.20. As this disparity was in the same direction for each subject at each delay interval, which has a binomial probability of 0.008, this model appears implausible.

The calculated results are inconsistent with the experimental data!

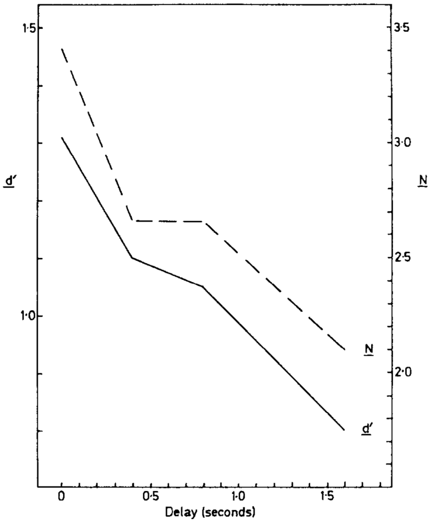
assumption

- in the S case (item included in the given set)
  - if the item corresponding to the probe has been retained, which has probability $p(t)$
  - subject will firmly answer: yes!
- in the $\bar{S}$ case (item not included in the given set)
  - if all 3 items retained
  - subject will firmly answer: no!
- otherwise
  - some items were lost in their BAS ...
  - subject will *guess*: yes...
  - (with probability of g)

probability of a correct response

$P(Y|S) = p(t) + (1 - p(t))g$ $P(Y|\bar{S}) = g(1 - p(t)^3)$

Results



- dashed line: the number of items in store N (calculated from this model)
- the mean value obtained for g: 0.22
- A sharp drop before 0.4s, then a flat 0.3s, and a slow drop afterward

Results

found similar small PR advantages, recognition of tones declined relative to a WR condition as the probe delay increased from $0 - 1.6$ s.

Compare two papers

- motivation:
  1. all research on the notion of BAS
  2. however, darwin's research was more focused on *echoic memory*, similar to the iconic memory for visual information
- methods:
  1. Darwin et al: PR & WR
  2. Treisman et al: PR & noncategorical stimuli & models
- results:
  1. Treisman et al: found *similar small PR advantages*, recognition of tones declined relative to a WR condition, to a near chance level as the probe delay increased from 0 to 1.6 s.
  2. What treisman found here is more closely to transient memory, treisman also thought the 4s darwin et al found was more likely a kind of *short-term memory*

Short and Long Auditory Stores [5]

Proposed Differences Between Short and Long Auditory Stores and Selected Relevant Evidence

Short auditory storage	Long auditory storage	Selected evidence
Duration: 150–350 ms	Duration: 2–20 s	Crowder, 1982b; Efron, 1970a, 1970b, 1970c; Plomp, 1964; data in Figure 1
Experienced as sensation	Experienced as memory	Broadbent, 1958; Efron, 1970a, 1970b, 1970c; Neisser, 1967; Plomp, 1964
Constant duration measured from event onset	Decay beginning at point of full stimulus resolution	Efron, 1970a, 1970b, 1970c; Massaro, 1970b, 1975; Pisoni, 1973
Storage of unanalyzed input?	Storage of partly analyzed input?	Cole, 1973; Massaro, 1972, 1976b; Wickelgren, 1965, 1966
Contains a spectral average weighted in favor of most recent input	Can contain a temporally structured sequence if one continuous stream	Blumstein & Stevens, 1980; Bregman, 1978; Dorman et al., 1975; Kewley-Port, 1983; Zwislocki, 1960, 1969
Interference is due to overwriting, and cannot be disinhibited	Interference is partial, depends upon target-mask similarity, and can be disinhibited	Crowder, 1978, 1982a; Kallman & Massaro, 1979, 1983

[5] Cowan, Nelson. "On short and long auditory stores." Psychological bulletin 96.2 (1984): 341.

Bibliographie I

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- Michel Treisman and AB Rostron. “Brief auditory storage: A modification of Sperling’s paradigm applied to audition.” In: *Acta Psychologica* 36.2 (1972), pp. 161–170.

Questions?